1.5 Part B: Independent OPITZ classification

1.5.1 Coding basis and methodology

The five-digit OPITZ form code $d_1d_2d_3d_4d_5$ describes the geometry of the rotational component. The classification below follows the digit descriptions in the APSD 26 CAPP/CNC OPITZ reference table supplied with this assignment. Each digit was first selected through a manual examination of the engineering drawing; the annotated figured are reproduced in Appendix **??**.

Table 1: Manual OPITZ decision worksheet based on the supplied APSD 26 digit table. The values represent the proposed manual code

Digit	Meaning	Value	Evidence and corresponding category
d_1	Part class	**1**	Rotational part; total length-to-maximum-diameter ratio $210/100 = 2.10$, within $0.5 < L/D < 3$.
d_2	External geometry	**5**	Outside diameter changes on either side of the flange; long right-hand section has an external thread. Selected category: stepped at both ends, thread.
d_3	Internal geometry	**2**	Axial through-bore, internal threaded region at the left and an enlarged bore at the right. Manual selection: stepped at one end, thread.
d_4	Plane surfaces	**0**	No distinct plane-surface machining was identified in the manual assessment; $R5$ transitions are fillets rather than plane surfaces. See note on possible end-face slot/keyway.
d_5	Auxiliary features	**2**	Six small axial holes, distributed on a pitch circle in the end view. Selected category: axial holes on pitch-circle diameter.

1.5.2 Questions and calculation

Q2. Geometric classification

The part is rotationally symmetric in its main turned geometry. The *overall* axial length is measured from the extreme left end to the extreme right end, $L = 210\,\text{mm}$. The 125 mm dimension shown above the long threaded region refers only to that portion and is *not* the overall part length. The largest relevant outside diameter includes the central flange, $D = 100\,\text{mm}$. Therefore,

$$\boxed{\frac{L}{D} = \frac{210\,\text{mm}}{100\,\text{mm}} = 2.10.} \tag{1}$$

According to Digit 1 of the supplied OPITZ table, rotational parts satisfying $0.5 < L/D < 3$ belong to part class $d_1 = 1$.

Q3. Five-digit form code

Digit 1 — Part class ($d_1 = 1$). The full axial length and the maximum flange diameter give $L/D = 2.10$. The selected reference row is "Rotational parts: $0.5 < L/D < 3$."

Digit 2 — External geometry ($d_2 = 5$). The component has changes in external diameter between the left cylindrical end, the large flange, and the long right-hand shaft. The long shaft displays an external thread. The two $R5$ annotations mark rounded shoulders (fillets), not necessarily separate grooves. The applicable reference-table entry is "Stepped both ends — Thread."

Digit 3 — Internal geometry ($d_3 = 2$; provisional). Section A–A shows a continuous axial opening, a dark threaded-looking region near the left opening, and a larger-diameter recess at the

right opening (approximately 40 mm long). The manual worksheet classified this as "Stepped on one end — Thread" and the choice is $d_3 = 2$.

Digit 4 — Plane-surface machining ($d_4 = 0$; provisional). No special plane-surface machining was identified in the manual worksheet. In particular, the $R5$ rounded transitions are fillets and do not by themselves constitute a planar milling operation. The working category is "No Surface Machining.".

Digit 5 — Auxiliary holes ($d_5 = 2$). The circular end view indicates six small holes distributed around the center. The holes are oriented parallel to the main axis and are arranged along a pitch circle rather than radially through the shaft. This matches the supplied Digit 5 entry "Axial on Pitch Circle Diameter."

The manual five-digit form code is therefore

$$\boxed{\mathbf{d_1 d_2 d_3 d_4 d_5 = 15202} \quad \text{(proposed manual classification)}}. \tag{2}$$

The potentially competing interpretations for d_3 and d_4 should be resolved against the complete feature definitions and original CAD model before treating the code as final.

Supporting manual evidence

Figures 2, 3, 4 and 6 reproduce the annotated drawing with handwritten reasoning from the independent manual analysis. The drawings show the length/diameter selection and external shape, followed by internal geometry, plane surfaces, and axial-hole pattern.

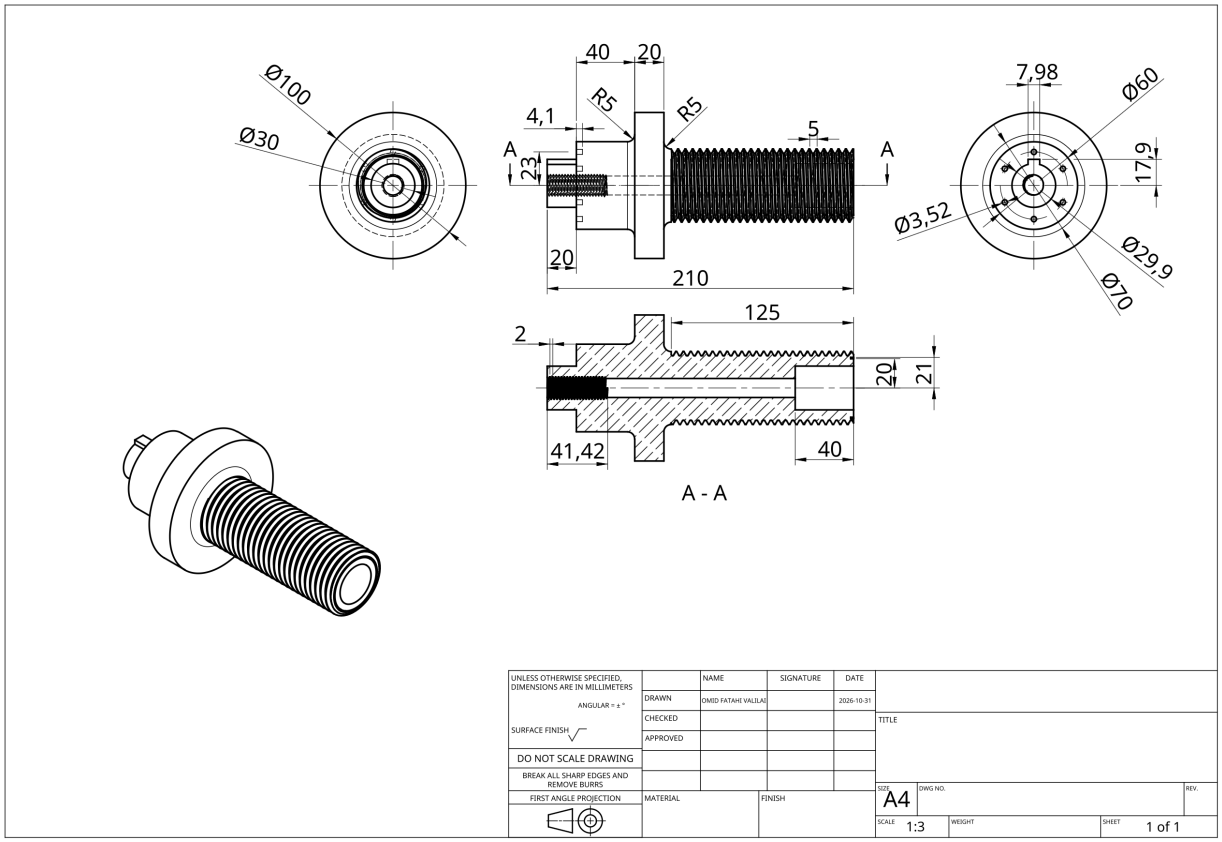


Figure 1: Original Engineering drawing of the Part

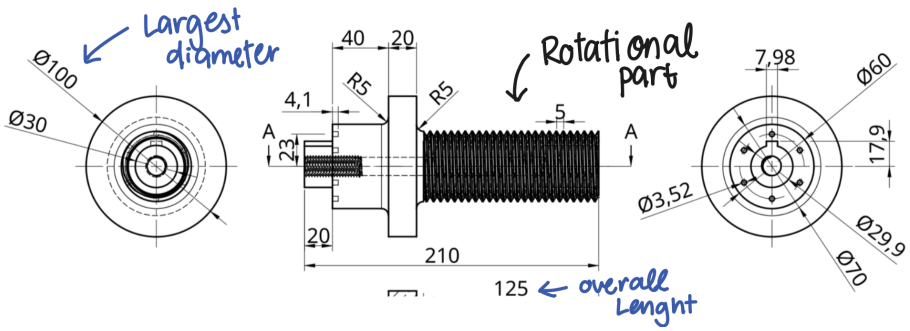


Figure 2: Annotated image of the Part's Left Side view, Front View and Right Side view for digit 1 analysis

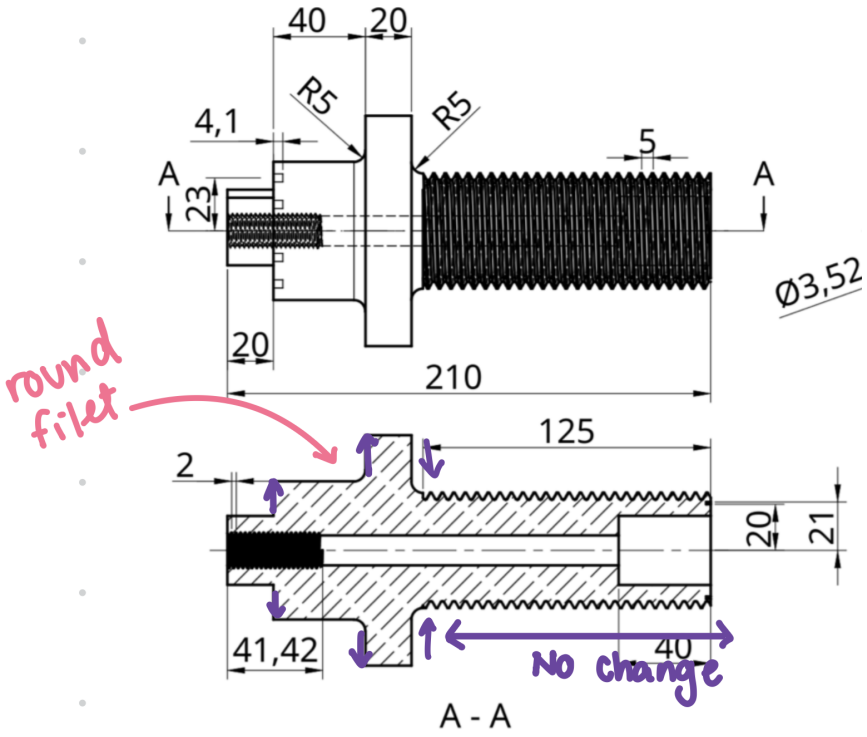


Figure 3: Annotated image of the Part's Front View and Cross section A for digit 2 analysis

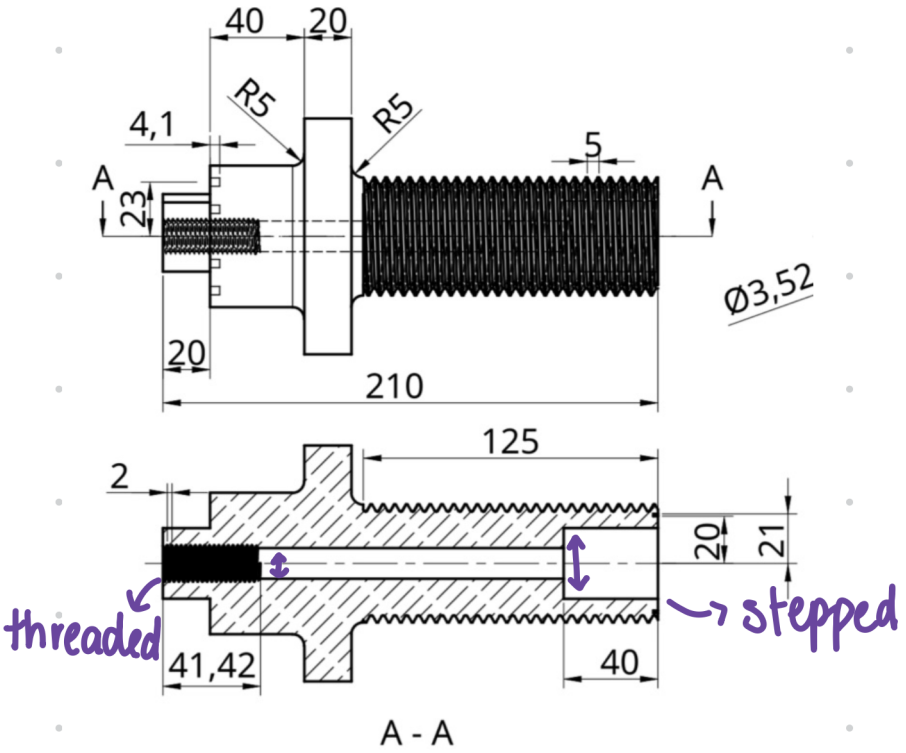


Figure 4: Annotated image of the Part's Front View and Cross section A for digit 3 analysis

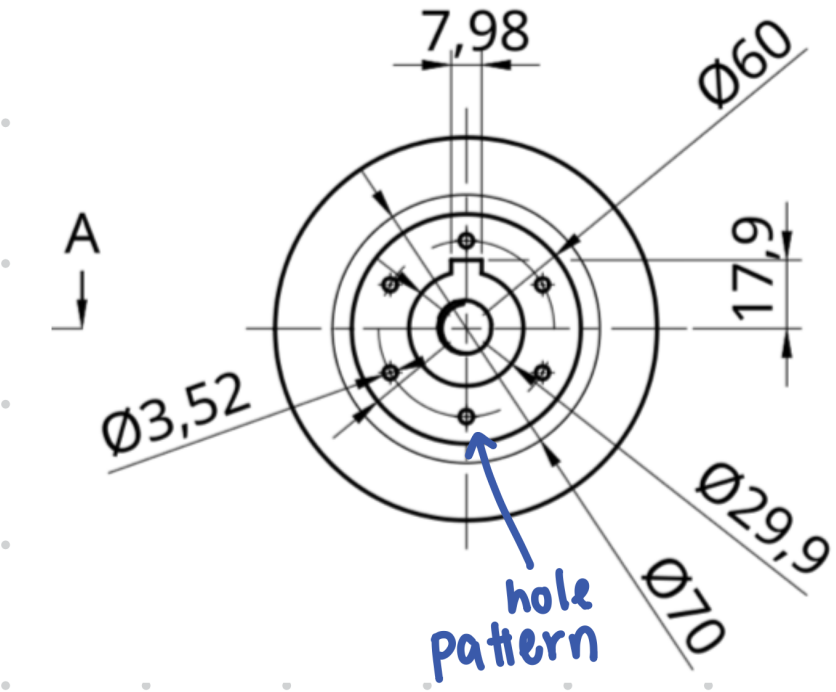


Figure 5: Annotated image of the Part's Left Side view for digit 5 analysis

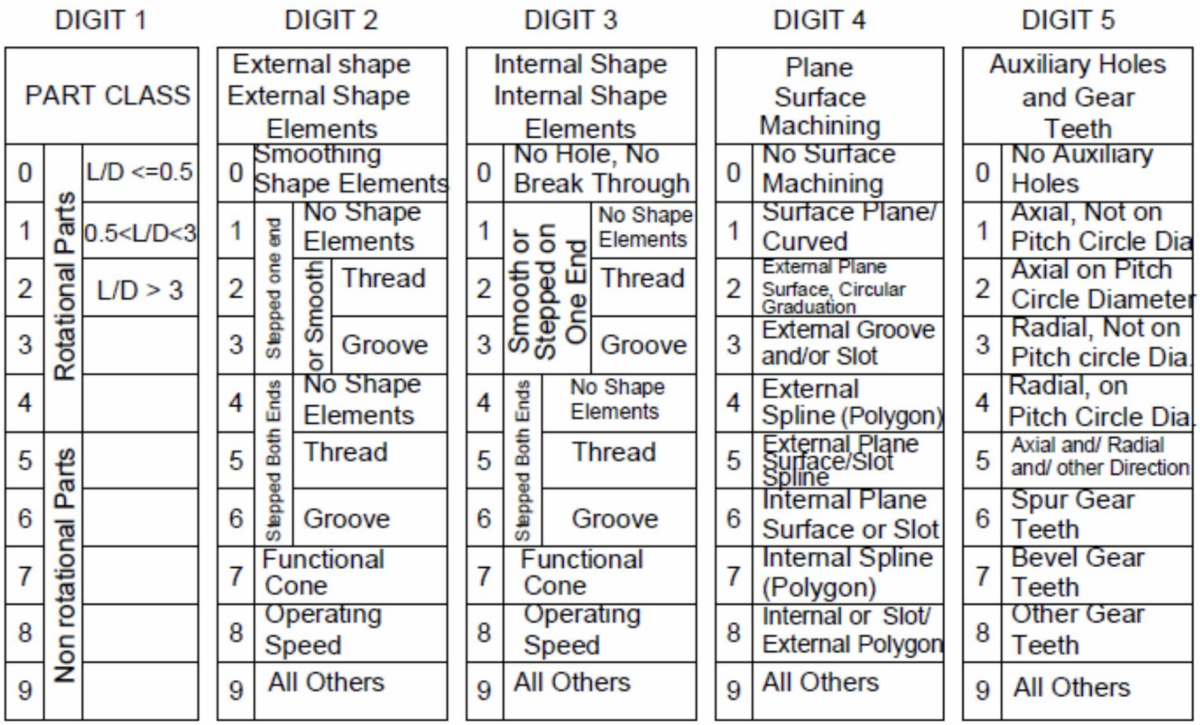


Figure 6: OPITZ code Table provided in the APSD26 class material

Manufacturing-assumption table

Table 2: Explicit planning assumptions, not specifications verified by the drawing.

Item	Working assumption for process planning
Material	Machinable metallic stock.
Initial stock	Round bar or near-net cylindrical blank sufficiently large to accommodate the flange diameter.
Manufacturing processes	Turning for coaxial diameters and shoulders; threading for the external shaft and tapping for the internal threading.
Accuracy/finish	General tolerances and surface finish, check the pattern holes around the pitch-circle.
Inspection	Verify the 210 mm overall length, 100 mm flange diameter, bore sizes, thread dimensions, and hole location/pitch circle.

Short proposed route sheet

Table 3: Illustrative route, contingent on the assumptions above and the verified CAD features.

Step	Operation	Purpose
1	Prepare stock	Cut and secure a suitable round blank.
2	Turn external features	Machine coaxial diameters, flange, shoulders and specified fillets.
3	Drill and bore	Establish the axial opening and right-hand counterbore.
4	Tapping	Create the internal thread in the left end of the drilling
5	Cut threads	Machine the long outside thread.
6	Drill auxiliary holes	Locate and produce the six axial holes on the specified pitch circle.
7	Finish and inspect	Deburr; inspect dimensions, hole pattern, bores and threads.